

Topics in CS: Problem Set 6

Due date: December 14, 2025.

Question 1. (20 points) Prove using the Miller-Rabin test that the Charnichael numbers 1105 and 1729 are composites.

Please write down the steps of your calculations; you can use a basic calculator for the task.

Question 2. (25 points) In class we stated that the fraction of strong witnesses in Miller-Rabin test is at least $3/4$ (note that there was a typo in the slide). In this question, we will prove a weaker statement.

Suppose that N is an odd composite number that is not a prime power (i.e., not of the form $N = p^t$ for some prime p and $t > 1$).

1. Prove that for every $k \in \mathbb{N}$ the set $S_k = \{x \in \mathbb{Z}_N^* \mid x^k \equiv \pm 1 \pmod{N}\}$ is a subgroup of $\mathbb{Z}_N^*$.
2. Denote $N = ab$ for some $a, b \in \mathbb{Z}$ such that $a, b > 1$ and $\gcd(a, b) = 1$ (here we rely on the assumption that N is not a prime power). Suppose that for some $k \in \mathbb{N}$ there exists $x \in S_k$ such that $x^k \equiv -1 \pmod{N}$.
 - (a) Prove that there exists $y \in \mathbb{Z}_N^*$ such that $y \equiv x \pmod{a}$ and $y \equiv 1 \pmod{b}$. (Hint, use Bezout's lemma.)
 - (b) Prove that for the element y from the previous item, it holds that $y \in \mathbb{Z}_N^* \setminus S_k$.
3. Denote $N = 2^r u$ where u is odd, and

$$\ell = \max\{0 \leq j \leq r \mid \exists x \in S_{2^j u} \text{ such that } x^{2^j u} \equiv -1 \pmod{N}\}.$$

Prove that ℓ is well defined, i.e., that there exists j such that $0 \leq j \leq r$ and $x^{2^j u} \equiv -1 \pmod{N}$.

4. Prove that if $x \in \mathbb{Z}_N^*$ is not a strong witness for the non-primality of N , then $x \in S_{2^\ell u}$.
5. Conclude that at least half of the elements in $\{1, \dots, N-1\}$ are string witnesses.

Question 3. (25 points) In this question you will attempt to create your own primality test!

1. We will start by looking at the composite number 15 and the prime number 13.
 - (a) Create a table where in the first column you write every element $a \in \mathbb{Z}_{15} \setminus \{0\}$ and in the second column $a^7 \pmod{15}$.
 - (b) Create a similar table where in the first column you write every element $a \in \mathbb{Z}_{13} \setminus \{0\}$ and in the second column $a^6 \pmod{13}$.
 - (c) Create another table where in the first column you write every element $a \in \mathbb{Z}_{13} \setminus \{0\}$ and in the second column $x^2 \pmod{13}$.
2. Do you see a pattern? If so, repeat the process with the composite number 9 and the prime number 7.
3. Design a primality test according to your findings.

You do not need to formally analyze the running time or the success probability, but specify under which assumptions your primality test works.

Question 4. (30 points)

1. Implement the algorithm Miller-Rabin.

Please use the code you implemented in previous assignments.

2. Implement the algorithm SamplePrime.

Sample 10 random primes with bit-length $n = 512$, using accuracy parameter $k = 40$.

For each such prime, count how many attempts are needed. What is the difference between the average of these attempts and the expected number of attempts from the prime number theorem?

Good luck!